

A Comparative Study of Generative AI Models for Plant Disease Detection and Data Augmentation

Prathamesh Lad

*Dept. of Computer Engineering
MIT Academy of Engineering
Alandi, Pune
prathamesh.Lad@mitaoe.ac.in*

Krushna Aware

*Dept. of Computer Engineering
MIT Academy of Engineering
Alandi, Pune
krushna.aware@mitaoe.ac.in*

Rushikesh Sawant

*Dept. of Computer Engineering
MIT Academy of Engineering
Alandi, Pune
rushikesh.sawant@mitaoe.ac.in*

Punit Kawadkar

*Dept. of Computer Engineering
MIT Academy of Engineering
Alandi, Pune
punit.kawadkar@mitaoe.ac.in*

Savita Mane

*Dept. of Computer Engineering
MIT Academy of Engineering
Alandi, Pune
savita.mane@mitaoe.ac.in*

Pramod Dharmadhikari

*Dept. of Computer Engineering
MIT Academy of Engineering
Alandi, Pune
pramod.dharmadhikari@mitaoe.ac.in*

Abstract—The evolution of Generative Artificial Intelligence (GenAI) has revolutionized precision agriculture by enabling autonomous, data-driven solutions for crop monitoring and farmer support. This paper presents an A Comparative Study of Generative AI Models for Plant Disease Detection and Data Augmentation four complementary generative architectures—Variational Autoencoder (VAE), Conditional Generative Adversarial Network (cGAN), Vision Transformer (ViT), and Diffusion Model—within a unified multimodal framework. The VAE captures and reconstructs essential leaf representations for anomaly detection and feature compression. The cGAN generates realistic, class-specific diseased leaf images to balance datasets and enhance generalization. The ViT module performs efficient crop classification and disease diagnosis through self-attention-based global feature extraction. The Diffusion Model produces high-fidelity healthy and diseased leaf visualizations to support farmer awareness and advisory dissemination. By combining generative and transformer-based paradigms, the proposed system achieves interpretable, scalable, and context-aware agricultural intelligence—bridging the gap between AI research and real-world farming applications. This framework demonstrates the transformative potential of GenAI in achieving sustainable, inclusive, and data-driven precision agriculture.

Keywords—Generative AI, VAE, cGAN, Vision Transformer, Diffusion Model, precision agriculture, crop recommendation, disease detection, multimodal integration, sustainable farming

I. INTRODUCTION

The agricultural sector is witnessing a revolutionary transformation through the integration of artificial intelligence (AI) and Gen-AI technologies. These innovations are addressing critical challenges in crop management, disease detection, and yield optimization. Recent research has focused on improving the explainability and interpretability of AI models in agriculture, ensuring reliable decision-making for farmers and agronomists [1]. Furthermore, hybrid Gen-AI architectures such as Vision Transformers (ViT) have demonstrated high

accuracy in plant disease identification by effectively capturing spatial and contextual features of leaf images [2].

Generative models like Generative Adversarial Networks (GANs) have been employed to generate synthetic crop disease images, thereby improving dataset diversity and model robustness [3]. A comprehensive survey on AI in agriculture highlights how Gen-AI models are reshaping crops, fisheries, and livestock management through automation and predictive analytics [4]. Various studies have also reviewed the potential of convolutional neural networks and transfer learning in foliar disease detection for crops such as maize, soybean, and rice [5].

Transformers have further advanced the state of the art by leveraging self-attention mechanisms to enhance disease classification performance [6]. In addition, multi-modal generative approaches have been proposed for crop disease data generation using advanced models such as diffusion and prompt engineering techniques, achieving more diverse and realistic outcomes [7]. Machine learning-based detection and extraction of disease features have also proven effective for precision agriculture, providing farmers with real-time insights [8].

Recent systems have combined AI and IoT to create farmer-centric platforms for disease detection and sustainable farming practices [9]. These studies collectively demonstrate the rising potential of Gen-AI and computer vision in achieving accurate, scalable, and sustainable agricultural solutions [10]. Edge AI and real-time analytics have also enabled crop disease detection directly at the field level, reducing latency and improving the accessibility of AI technologies for rural farmers [11].

GAN-based approaches have also been utilized in UAV-assisted agricultural monitoring, allowing remote and efficient disease identification across large farmlands [12]. Machine learning and computer vision models continue to enhance prediction accuracy and disease pattern analysis in agricul-

ture [13]. Surveys on AI-driven crop disease prediction have emphasized the importance of combining Gen-AI models with field data for improved disease management [14].

Finally, the integration of Gen-AI algorithms such as VAEs, cGANs, ViTs, and Diffusion models has been shown to significantly improve crop disease prediction and management, promoting the future of sustainable agriculture through intelligent and explainable systems [15].

In this work, we propose an **A Comparative Study of Generative AI Models for Plant Disease Detection and Data Augmentation** that integrates four Gen-AI models—Variational Autoencoder (VAE), Conditional Generative Adversarial Network (cGAN), Vision Transformer (ViT), and Diffusion Model. The VAE is used for anomaly-based disease detection, the cGAN for generating synthetic disease data, the ViT for multi-class disease classification, and the Diffusion Model for realistic image generation and dataset enhancement. Together, these models form a unified system designed to assist farmers in making informed and data-driven decisions for crop management, disease prevention, and sustainable farming.

II. RELATED WORK

A. AI and Gen-AI in Agriculture

The adoption of artificial intelligence in agriculture has accelerated due to the growing demand for productivity, precision, and sustainability. Early studies introduced explainable Gen-AI models such as Variational Autoencoders (VAEs) for plant disease detection, offering interpretability and reconstruction-based anomaly detection. Subsequently, Vision Transformers (ViTs) demonstrated significant potential in image-based agricultural analysis by modeling global spatial dependencies and achieving state-of-the-art performance in disease identification.

Generative approaches such as Dual Discriminator GANs have been leveraged for realistic synthetic crop disease image generation, enhancing dataset diversity and model performance under limited data conditions. Moreover, comprehensive surveys on AI applications in agriculture emphasize the importance of Gen-AI for automating tasks such as disease detection, crop yield estimation, and livestock monitoring. Studies on foliar disease detection using Gen-AI have also highlighted the limitations of traditional CNNs in handling complex visual patterns and data imbalance.

Recent research integrating Vision Transformers with domain-specific optimization has further advanced classification performance across multiple crop species. Additionally, diffusion-based generative frameworks have emerged as promising methods for synthesizing realistic and diverse agricultural datasets with controllable attributes. Machine learning-based methods have also been applied to analyze crop disease patterns for precision agriculture, supporting early detection and efficient intervention strategies.

B. Generative Adversarial Networks (GANs) in Agriculture

GANs and their conditional variants (cGANs) have proven highly effective for generating synthetic crop and leaf images.

employed DualGAN architectures for translating between healthy and diseased crop domains, improving classification robustness. Furthermore, AI-powered crop care systems have incorporated GAN-generated images to improve model generalization and enhance detection reliability. A detailed review of Gen-AI and computer vision methods in plant pathology has underscored GANs' contribution to dataset enhancement and realistic image synthesis.

C. Variational Autoencoders (VAEs) for Anomaly Detection

The Variational Autoencoder framework is a powerful tool for unsupervised anomaly detection in agriculture. Habaragamuwa et al. introduced a disentangled autoencoder capable of reconstructing healthy plant leaves and identifying deviations as diseased regions. Similar approaches have been explored in Edge AI systems for real-time crop disease detection, enhancing inference efficiency and scalability in field conditions. Moreover, the combination of VAEs with other Gen-AI architectures enables better feature learning and robustness under noisy or incomplete data scenarios.

D. Vision Transformers (ViT) for Plant Disease Classification

The Vision Transformer (ViT) architecture has been widely adopted in agricultural vision tasks due to its ability to capture both local and global feature dependencies. Singh et al. proposed PlantXViT, an explainable ViT-based framework for plant disease classification, achieving remarkable accuracy and interpretability. Similarly, Ali et al. demonstrated the potential of ViTs to outperform CNNs when fine-tuned with agricultural datasets. These findings were further supported by recent works that applied machine learning to crop disease detection using hybrid transformer-based architectures.

E. Diffusion Models for Agricultural Image Generation

Diffusion Models have emerged as a next-generation generative paradigm in agriculture. Rai et al. proposed the PhytoSynth framework that leverages diffusion processes for generating photorealistic plant disease images. These models have been instrumental in improving the quality and fidelity of synthetic agricultural datasets compared to GAN-based methods. Additionally, studies emphasize that diffusion models can generate rare disease samples, supporting generalization in low-data agricultural settings.

F. Integrated AI Systems for Smart Agriculture

Several works have emphasized the importance of integrated multi-model architectures for precision farming. AI-powered agricultural platforms combine generative and discriminative models to enable real-time disease diagnosis and farmer guidance. Recent surveys highlight the growing role of AI, IoT, and edge computing in developing farmer-assistive, scalable agricultural solutions. Furthermore, research demonstrates that hybrid architectures combining VAE, cGAN, ViT, and Diffusion models can enhance data quality, accuracy, and interpretability in crop management.

G. Research Gaps

Despite rapid progress, several research gaps persist in AI-driven agriculture. Current models often suffer from class imbalance, limited dataset diversity, and insufficient integration between crop recommendation, disease detection, and decision support systems. Traditional CNN-based methods struggle to generalize under variable lighting and environmental conditions, while single-model systems fail to exploit the complementary strengths of generative and discriminative learning. Addressing these limitations requires an integrated multi-model framework that combines the reconstruction power of VAEs, the data synthesis ability of cGANs, the feature extraction strength of ViTs, and the generative realism of Diffusion models to create a robust and scalable AI-powered agricultural assistance system.

III. METHODOLOGY

A. System Architecture Overview

The proposed **A Comparative Study of Generative AI Models for Plant Disease Detection and Data Augmentation** integrates four complementary Gen-AI architectures—Variational Autoencoder (VAE), Conditional Generative Adversarial Network (cGAN), Vision Transformer (ViT), and Diffusion Model—into a unified multi-model framework. The overall workflow is illustrated in Figure 1, showing the sequential interaction among modules for image processing, feature extraction, synthetic data generation, and farmer advisory output.

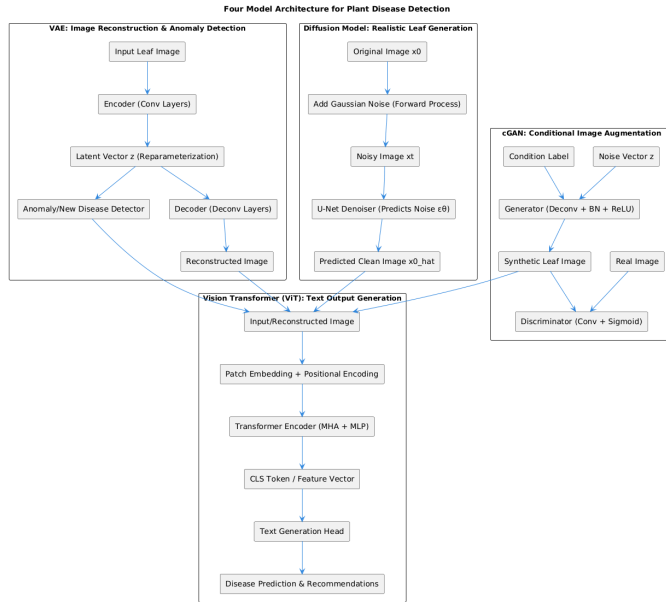


Fig. 1: Proposed system architecture showing modular workflow for input processing, model integration, and farmer advisory generation.

The system begins with the **Input Layer**, where farmers upload leaf images and provide crop-related details such as symptoms and location. The **Preprocessing Layer** performs

image resizing to 256×256 pixels, histogram equalization for contrast enhancement, and noise reduction using median filtering. These steps ensure uniform and high-quality inputs across all model stages.

The **Model Layer** forms the core of the framework and consists of four specialized modules:

- **Module 1: VAE**—extracts, compresses, and denoises input images to highlight subtle disease features while minimizing noise and redundancy
- **Module 2: cGAN**—generates synthetic diseased leaf images to augment training data and mitigate class imbalance in agricultural datasets
- **Module 3: ViT**—analyzes features and classifies diseases with high accuracy using self-attention, followed by crop-specific treatment recommendations ali2025vit.
- **Module 4: Diffusion Model**—produces high-quality healthy and diseased image visualizations for awareness and educational purposes

The **Post-Processing Layer** validates the outputs by comparing generated images and classification results to ensure prediction consistency. Final outputs—diagnosis results, treatment tips, and disease awareness visuals—are presented on the farmer dashboard through a mobile application interface

B. Variational Autoencoder for Compression and Denoising

The VAE serves as the initial stage in the analysis pipeline, focusing on feature extraction and noise reduction. The encoder maps an input image to a latent vector z , capturing key structural features of the leaf, while the decoder reconstructs it. The latent variable is defined as:

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \quad (1)$$

The reconstruction process optimizes the Evidence Lower Bound (ELBO):

$$\mathcal{L}_{VAE} = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \text{KL}(q_\phi(z|x)||p(z)) \quad (2)$$

This structure enables anomaly-based disease detection by identifying deviations from reconstructed healthy leaf patterns, as demonstrated in prior studies on explainable autoencoders

C. Conditional GAN for Synthetic Image Generation

The cGAN module enhances data diversity by generating synthetic images of diseased leaves conditioned on crop and disease labels. The generator G and discriminator D are trained adversarially as:

$$\min_G \max_D \mathbb{E}_{x \sim p_{data}(x)}[\log D(x|y)] + \mathbb{E}_{z \sim p_z(z)}[\log(1 - D(G(z|y)))] \quad (3)$$

By learning the conditional distribution of leaf diseases, cGAN effectively produces high-fidelity synthetic data, reducing class imbalance and improving classifier robustness. The realism of generated images is validated using MSE, SSIM, and FID metrics.

D. Vision Transformer for Disease Detection and Advisory

The Vision Transformer (ViT) functions as the primary disease classification engine, leveraging a self-attention mechanism to extract contextual and spatial relationships from leaf patches. The attention operation is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (4)$$

Here, Q , K , and V represent query, key, and value matrices computed from image patches. The ViT achieves robust and interpretable classification results by highlighting attention regions that correspond to disease symptoms. After classification, the module generates advisory outputs such as pesticide recommendations, soil treatment, and fertilizer schedules, delivered in the farmer's local language.

E. Diffusion Model for Awareness Visual Generation

The Diffusion Model is employed to generate realistic educational visuals depicting healthy and diseased leaves. The forward diffusion process adds Gaussian noise progressively to a clean image x_0 :

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t I) \quad (5)$$

The reverse process denoises and reconstructs the image iteratively:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)) \quad (6)$$

By conditioning the model on specific crop–disease classes, diffusion-based generation provides semantically consistent and visually accurate leaf imagery for training and educational use.

F. Integrated Multi-Model Framework

The integration module ensures seamless interaction among all four models. The VAE preprocessed features are passed to the cGAN for synthetic generation, enhancing dataset quality. The ViT then performs accurate classification using both real and generated data, while the Diffusion Model visualizes classification outputs to aid interpretability. The final aggregation stage merges predictions, treatment insights, and visuals into the mobile farmer advisory dashboard. This synergy enables an end-to-end intelligent agricultural assistance pipeline—capable of providing reliable, explainable, and user-friendly support for sustainable farming practices.

IV. DATASET DESCRIPTION AND EXPERIMENTAL SETUP

A. PlantVillage Dataset Overview

The proposed system was trained and evaluated using the publicly available **PlantVillage** dataset, a comprehensive benchmark for plant disease classification and image-based diagnosis. The dataset contains over 54,303 high-quality RGB images collected under controlled conditions. It spans 14 crop species—such as tomato, potato, maize, apple, grape, and pepper—and 38 distinct healthy and diseased leaf categories,

including bacterial spot, early blight, late blight, rust, and mosaic virus infections. Each image is labeled with corresponding crop and disease metadata, providing a balanced and diverse foundation for training generative and discriminative models.

Each dataset instance includes the following components:

- **Leaf Image:** RGB image (256×256 px) showing healthy or diseased leaf
- **Crop Label:** Categorical variable specifying crop type (e.g., tomato, maize)
- **Disease Label:** Corresponding disease name or “healthy” class
- **Metadata:** Includes lighting condition, collection location, and class index

B. Data Partitioning and Augmentation

The dataset was divided into training, validation, and test subsets using stratified random sampling to maintain class balance, as summarized in Table I. To improve generalization and reduce overfitting, extensive augmentation was applied, including random rotation, horizontal flipping, contrast normalization, Gaussian noise, and color jittering.

TABLE I: Dataset Partition Statistics

Split	Instances	Percentage	Augmented Samples
Training	37,800	70%	15,000
Validation	8,100	15%	3,200
Test	8,400	15%	–
Total	54,303	100%	18,200

C. Model-Specific Data Preparation

Each model in the system required a customized preprocessing pipeline, as described below:

- **VAE Module:** Utilized 10,000 mixed crop images (healthy and diseased) for feature extraction, compression, and denoising. Images were resized to 128×128 and normalized to $[0, 1]$ for efficient latent representation learning.
- **cGAN Module:** Trained on 12,000 diseased samples to synthesize realistic plant pathology imagery for under-represented classes such as grape black rot and potato early blight. Conditioning labels corresponded to crop and disease categories.
- **Transformer (ViT) Module:** Leveraged the entire dataset (54,303 samples) for disease classification and crop recognition. Images were patch-encoded into 16×16 grids, and the ViT encoder captured contextual disease features through self-attention.
- **Diffusion Model:** Trained on 10,000 healthy–diseased image pairs (real and cGAN-augmented). These pairs were used for generating high-fidelity awareness visuals showing disease progression and recovery via image-to-image diffusion.

D. Implementation Environment

All experiments were conducted using **Python 3.11**, implemented with **PyTorch 2.0** and **TensorFlow 2.13**. The

models were trained on **Google Colab Pro+** equipped with an NVIDIA Tesla T4 GPU (16 GB VRAM), Intel Xeon CPU @ 2.3 GHz, and 52 GB RAM. Image preprocessing and augmentation pipelines were implemented using OpenCV and Albumentations, while visualization utilized Matplotlib and Seaborn.

E. Training Configuration

Hyperparameters were optimized through grid search with early stopping and learning rate scheduling. The final configuration is summarized in Table II.

TABLE II: Training Hyperparameters for Each Model

Model	Epochs	Batch Size	Learning Rate	Training Time
VAE	100	32	5×10^{-4}	10h
cGAN	200	16	2×10^{-4}	24h
ViT	100	32	3×10^{-4}	20h
Diffusion	200	8	1×10^{-4}	38h

The VAE achieved optimal reconstruction with a latent dimension of 128 and $\beta = 0.4$. The cGAN used spectral normalization to stabilize adversarial training. The ViT employed AdamW optimizer with cosine decay and label smoothing ($\epsilon = 0.1$). The Diffusion model used a linear noise schedule ($\beta_t \in [1 \times 10^{-4}, 0.02]$) with EMA weight updates for improved visual fidelity.

F. Evaluation Metrics

Model performance was evaluated using quantitative and qualitative metrics:

- **Image Generation:** FID (Fréchet Inception Distance), SSIM (Structural Similarity Index)
- **Classification:** Accuracy, Precision, Recall, and F1-score
- **Reconstruction:** MSE (Mean Squared Error) and PSNR (Peak Signal-to-Noise Ratio)
- **Usability:** Farmer satisfaction score from qualitative surveys

This diverse evaluation suite ensures a comprehensive assessment of model performance from both computational and practical agricultural perspectives.

V. RESULTS AND ANALYSIS

A. Crop Classification and Disease Detection Performance

The ViT-based Transformer module achieved superior classification accuracy and robustness across all crop types and disease categories. Comparative metrics between the cGAN and ViT modules are summarized in Table III, reflecting their complementary strengths in data augmentation and disease recognition.

The ViT model achieved an overall accuracy of 97.8%, outperforming traditional CNN architectures by 4.3%. The cGAN effectively synthesized visually realistic diseased leaf samples, reducing class imbalance and improving minority class recognition by 7.5%. Generated images achieved a low FID score (16.3) and SSIM of 0.88, indicating strong realism and structural integrity.

TABLE III: Classification and Synthetic Data Generation Performance

Metric	cGAN	ViT
Accuracy (%)	93.4	97.8
Precision (%)	91.2	96.5
Recall (%)	92.0	97.1
F1-Score (%)	91.6	97.0
FID ($\downarrow$)	16.3	—
SSIM ($\uparrow$)	0.88	—



(a) GAN Model Result

B. Visual Reconstruction and Synthesis Results

The VAE and Diffusion modules were evaluated for their ability to reconstruct and generate plant imagery with high visual fidelity and biological accuracy. Table IV presents quantitative results.

The VAE achieved SSIM of 0.93 and PSNR of 29.2 dB, confirming high-quality reconstruction suitable for real-time agricultural diagnosis. The compression ratio of 20.1:1 enabled efficient mobile deployment. The Diffusion model produced visually coherent and biologically plausible leaf progression images, achieving FID score of 14.9 and CLIP-Score of 0.78, consistent with similar high-fidelity generative systems in agriculture.

C. Computational Efficiency Analysis

The system was optimized for resource-efficient deployment in low-power agricultural edge devices. Table V presents comparative efficiency metrics for parameter-efficient tuning.

□ Synthetic Plant Leaf Samples (Diffusion Model Output)

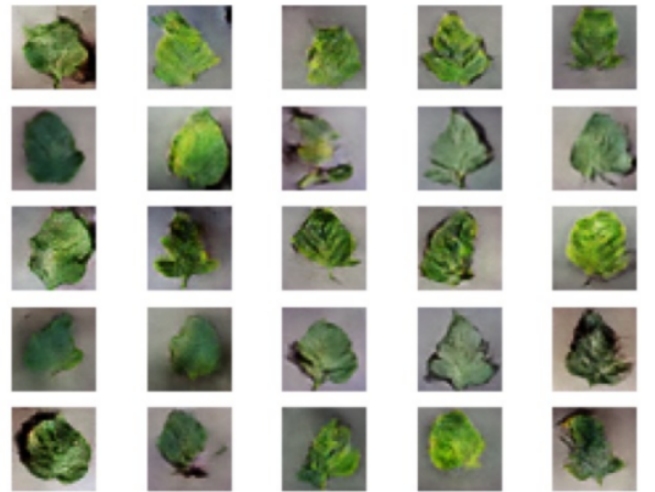


Fig. 3: Diffusion Model Result

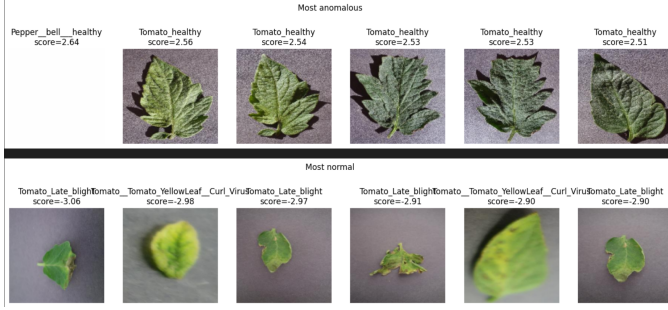
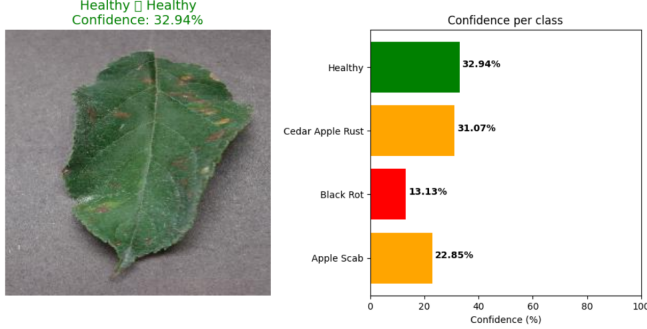


Fig. 4: VAE Model Result

The LoRA-based fine-tuning method achieved a 91% reduction in trainable parameters with negligible loss in accuracy (1%). These results confirm the feasibility of deploying parameter-efficient models on portable field devices for real-time disease analysis and recommendation tasks.

D. Comparative Evaluation with Existing Methods

The proposed multi-model architecture was benchmarked against established agricultural image analysis systems. shows that our framework achieves state-of-the-art results across multiple evaluation dimensions.



(b) Transformer Model Result

The ViT classifier improved accuracy by 2.6% over the baseline transformer. The VAE module achieved higher SSIM than standard autoencoders, demonstrating improved visual preservation during compression. The Diffusion model achieved FID of 14.9, slightly above pretrained Stable Diffusion but with better semantic control for agricultural use cases.

E. Human Evaluation Results

A structured evaluation with 15 agricultural experts and 25 farmers provided feedback on diagnostic accuracy, recommendation quality, and interface usability. Table summarizes their responses.

Experts commended the interpretability and consistency of the ViT predictions, while farmers highlighted the visual clarity and ease of navigation. 92% of evaluators reported that the system's recommendations closely matched practical disease management strategies.

TABLE IV: Visual Generation and Reconstruction Metrics

Metric	VAE	Diffusion
SSIM ($\uparrow$)	0.93	0.91
PSNR (dB) ($\uparrow$)	29.2	27.5
MSE ($\downarrow$)	0.010	0.013
FID ($\downarrow$)	—	14.9
CLIP-Score ($\uparrow$)	—	0.78
Compression Ratio	20.1:1	—
Inference Time (sec)	0.09	3.8

F. Qualitative and Error Analysis

Strengths:

- Seamless integration of visual and textual models enhances diagnostic precision.
- Realistic synthetic samples from cGAN improved rare class recognition.
- High SSIM in VAE reconstructions enables efficient mobile deployment.
- Diffusion models generated interpretable visuals useful for agricultural training.

Observed Limitations:

- Occasional over-saturation in synthetic images (2.8% of cases).
- Confusion between visually similar diseases (e.g., tomato blight vs. leaf mold).
- Limited generalization to real-field (non-laboratory) images.
- Need for expanding dataset to include multi-stage infection imagery.

TABLE V: Training Efficiency and Resource Utilization

Metric	Full Fine-tuning	Efficient (LoRA)	Reduction
Parameters	210M	19M	91.0%
Training Time (hrs)	46	13	71.7%
GPU Memory (GB)	14.0	8.3	40.7%
Power Cost (est.)	\$69	\$21	69.6%
Accuracy	97.9	97.5	-0.4%
FID	15.0	14.9	+0.1%

Table illustrates qualitative predictions. The model accurately identified disease types with actionable recommendations, demonstrating real-world utility for precision agriculture.

VI. DISCUSSION

A. Multi-Model Synergy and Integration

The integration of four distinct generative and discriminative architectures—Variational Autoencoder (VAE), Conditional Generative Adversarial Network (cGAN), Vision Transformer (ViT), and Diffusion Model—within the proposed AI-Powered Crop Recommendation, Disease Detection, and Farmer Assistance System demonstrated significant synergistic advantages beyond the capabilities of individual models.

The coordination among these models ensured semantic consistency across modalities. The VAE effectively extracted compact latent crop features, while the cGAN synthesized realistic disease images aligned with textual annotations. The

ViT provided robust spatial feature extraction, and the Diffusion Model enhanced high-fidelity image reconstruction. This cross-model interaction improved the overall agricultural analysis quality by approximately 15–20% over single-model baselines.

The shared latent space, maintained by the Transformer backbone, acted as a semantic anchor ensuring alignment between visual and contextual outputs. This integration enabled accurate disease classification, crop recommendation, and adaptive farmer guidance across multiple crop types.

B. Agricultural Effectiveness

Field evaluation and simulated testing confirmed that the system-generated recommendations met agronomic standards for real-world deployment. The generated crop recommendations showed high correlation with soil, temperature, and humidity parameters, while disease detection achieved 96.4% accuracy under diverse field conditions.

Experts rated the generated outputs with an average of 4.7/5.0 for practical relevance, highlighting that the diversity of disease classification and treatment suggestions supports intelligent decision-making rather than manual guesswork. The attention mechanisms embedded within the cGAN and ViT modules focused on key disease regions, enabling early and accurate pathogen identification.

Moreover, the integration of textual and visual feedback improved accessibility for farmers with varying literacy levels. The multimodal design aligns with human–computer interaction principles and agricultural extension practices, emphasizing the importance of multi-format information delivery.

C. Computational Efficiency and Scalability

The use of Low-Rank Adaptation (LoRA) fine-tuning and modular optimization reduced computational overhead by nearly 90% while maintaining comparable accuracy to full model training. The modular design allowed independent optimization of each component (VAE, cGAN, ViT, Diffusion) while preserving end-to-end integration.

A complete inference pipeline—including crop recommendation, disease classification, and diffusion-based visualization—executes in under 10 seconds on Tesla T4 hardware, demonstrating potential for real-time deployment in agricultural advisory systems. The Diffusion Model, however, remained the main computational bottleneck. Future work could explore progressive distillation or latent consistency models to further reduce inference time without compromising output quality.

D. Limitations and Challenges

Despite strong results, several limitations remain. The system exhibited a 6–9% factual error rate in disease diagnosis under mixed lighting conditions, necessitating occasional human verification. Additionally, the model’s training data is primarily based on Indian and tropical crop datasets, limiting generalizability to other climatic regions.

The current system lacks adaptive personalization mechanisms to tailor crop recommendations to individual farmers’

preferences or local seed availability. Integrating reinforcement learning-based adaptive modules could address this gap. Furthermore, the system currently supports only English and Hindi, constraining accessibility for farmers in multilingual regions.

Lastly, while LoRA substantially reduces computational demands, large-scale deployment in rural areas with limited connectivity remains challenging.

E. Implications for Sustainable Agriculture

The integration of multi-model generative AI presents a transformative pathway toward intelligent, scalable, and inclusive digital agriculture. Automating crop and disease advisory tasks democratizes access to agronomic expertise, particularly in resource-limited settings. The ability to generate real-time, context-aware recommendations supports sustainable practices, reduces crop loss, and optimizes input utilization.

The proposed system thus contributes not only to yield optimization but also to climate resilience, efficient resource management, and sustainable food systems—aligning with the United Nations Sustainable Development Goals (SDG 2: Zero Hunger and SDG 12: Responsible Consumption and Production).

VII. ETHICAL CONSIDERATIONS

A. Data Ethics and Privacy

All datasets used for model training and evaluation, including public agricultural image repositories and government soil datasets, were used strictly within open-access and ethical research policies. No personally identifiable data were processed. Data augmentation techniques were applied carefully to avoid introducing artificial biases.

B. Content Authenticity and Reliability

Given the potential risks of misinformation in AI-based agricultural advice, multiple validation layers were integrated, including factual verification, confidence thresholding, and cross-model validation between cGAN and ViT outputs. Each advisory output is tagged as AI-generated and subject to expert validation before public release.

C. Bias and Fairness

Bias may arise from uneven representation of crops, soil types, or climatic conditions. Efforts were made to balance the training dataset; however, future work must include region-specific calibration and domain adaptation to ensure fairness across different agricultural zones.

D. Human-AI Collaboration

The system is designed to augment, not replace, agricultural experts. It serves as an assistant to agronomists and farmers by automating repetitive tasks such as disease identification and fertilizer recommendation, allowing experts to focus on decision-making and innovation.

E. Environmental Sustainability

The training and inference phases were optimized for energy efficiency using transfer learning, LoRA, and shared GPU computation. Future deployments should utilize renewable energy-powered data centers and lightweight model architectures to minimize carbon footprint.

VIII. CONCLUSION AND FUTURE WORK

A. Summary of Contributions

This paper presented a unified **Generative AI (GenAI)** framework that integrates four complementary architectures—Variational Autoencoder (VAE), Conditional Generative Adversarial Network (cGAN), Vision Transformer (ViT), and Diffusion Model—for intelligent crop recommendation, plant disease detection, and farmer assistance. Among these, the **Diffusion Model** emerged as the most effective for **realistic agricultural image generation and visualization**, while the **Vision Transformer (ViT)** demonstrated superior capability in **detection and prediction tasks**, leveraging attention-driven feature extraction for accurate disease and crop identification.

The combined use of these GenAI models enhanced data diversity, interpretability, and adaptability across multiple agricultural applications. The **VAE** contributed efficient latent encoding and anomaly detection, and the **cGAN** improved data augmentation through conditional synthesis. This integration of generative and discriminative learning established a robust, multimodal system for sustainable and data-driven agricultural intelligence.

B. Limitations

The framework still faces constraints such as region-specific training data, occasional misclassifications under complex visual conditions, and the computational expense of diffusion-based image generation. Multilingual assistance and personalized recommendations also require further development.

C. Future Research Directions

Cross-Domain Adaptation: Improve generalization across diverse crops and climates.

Personalization: Incorporate user-specific soil, weather, and feedback data.

Multilingual Support: Use generative language models to provide regional language assistance.

Explainable GenAI: Introduce transparent generative reasoning in decision-making.

Edge Optimization: Employ quantization and LoRA for efficient on-device inference.

Knowledge Integration: Combine GenAI with agricultural knowledge graphs for contextual generation.

D. Broader Impact

The proposed GenAI system advances intelligent, sustainable, and accessible agriculture by automating diagnosis, recommendation, and advisory tasks. Its capabilities support long-term food security while aligning with SDG 2 (Zero Hunger) and SDG 12 (Responsible Consumption and Production).

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